

Thermoluminescence of MgO Chemically Modified with Nd and Li Ions, Obtained by Glycine-based Solution Combustion.



"El saber de mis hijos
hará mi grandeza"

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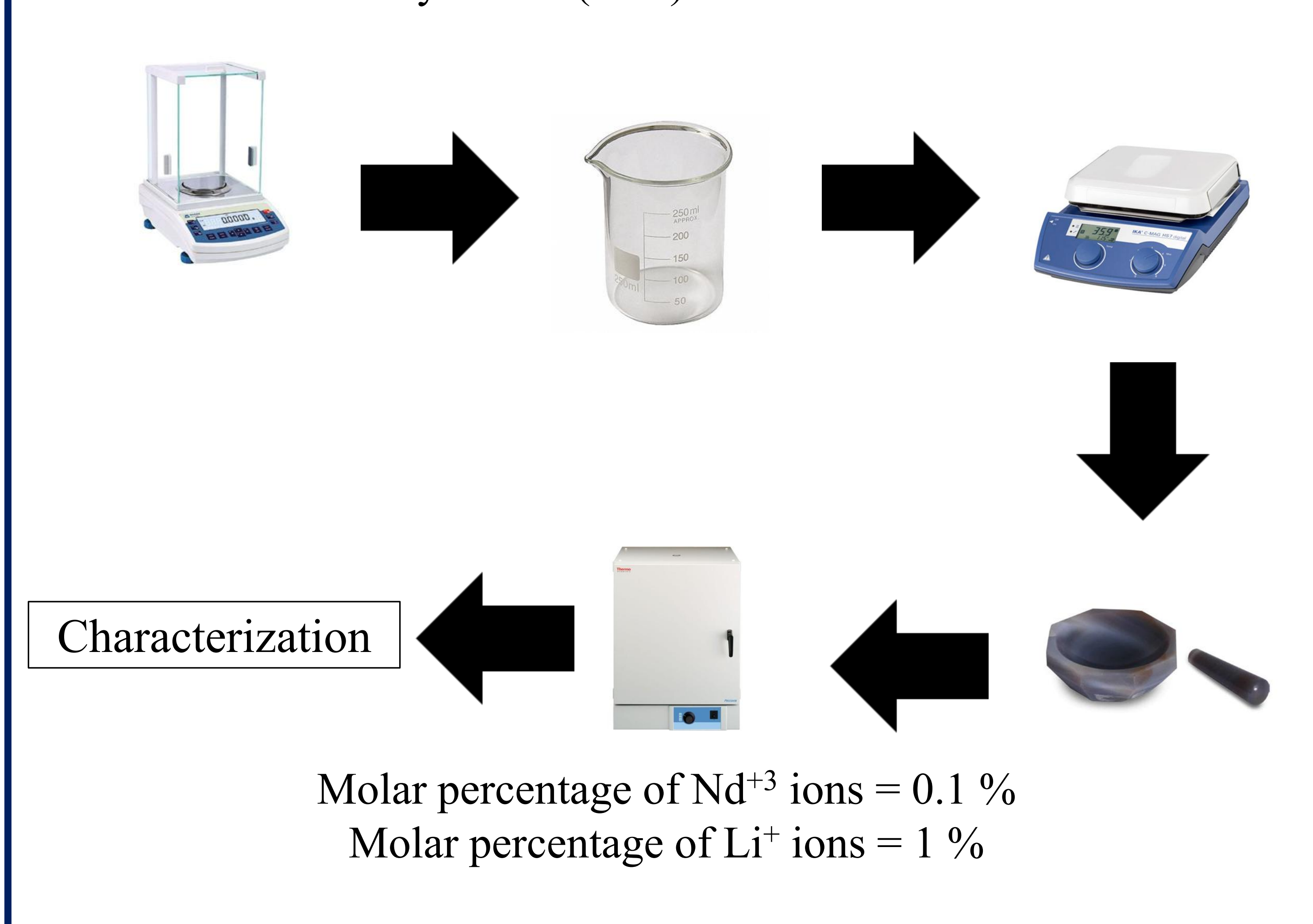
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INTRODUCTION

MgO is a very attractive material in many different applications, such as insulators in industrial cables, crucibles, and refractory materials. The combustion method has been successfully used in the preparation of technologically useful oxides. This oxide material, which is involved in many research topics, has been doped with transition metals and has been extensively studied to understand the formation of defects, the properties of these defects, and their role in optical and electrical properties, on the other hand, the thermally stimulated luminescence technique (TL), is widely accepted as a useful and reliable technique for the study of defects in insulating and semiconductor materials, but the most successful and most widely disseminated application of TL is in the field of radiation dosimetry.

MATERIALS AND METHOD

Solution combustion synthesis (SCS).



RESULTS

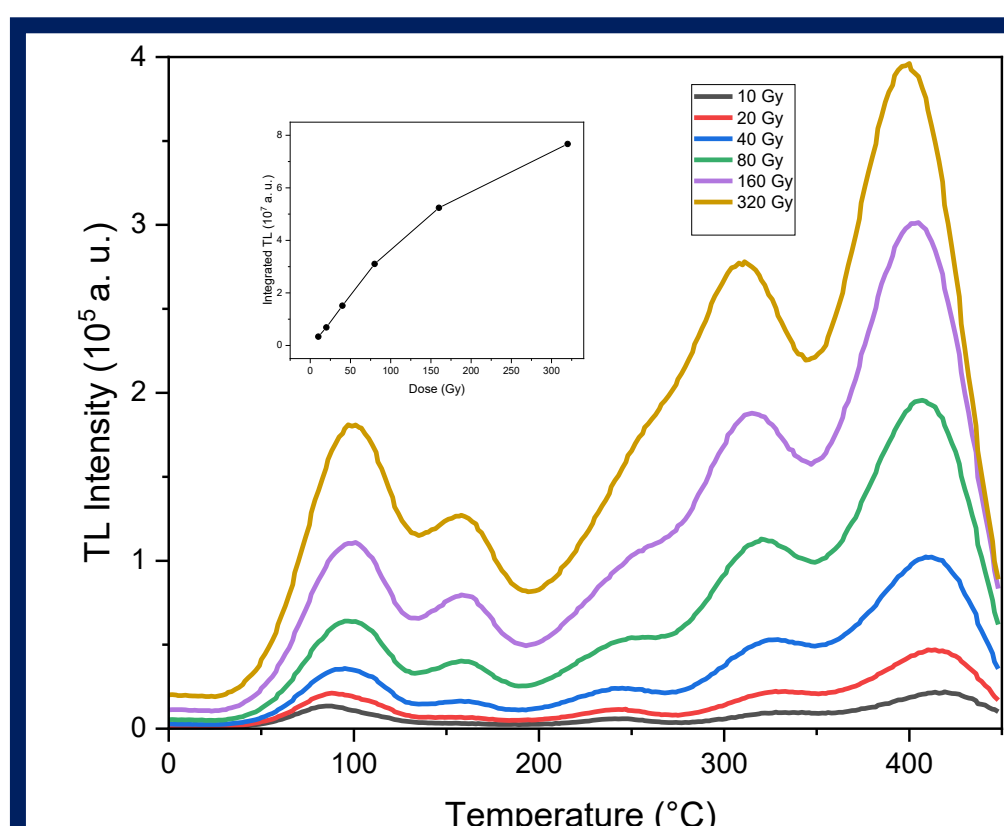


Figure 1. Evolution of thermoluminescent glow curves of MgO powder glycine-based chemically modified with Nd and Li, previously exposed to different doses of beta radiation and dose response showed in the inner graph.

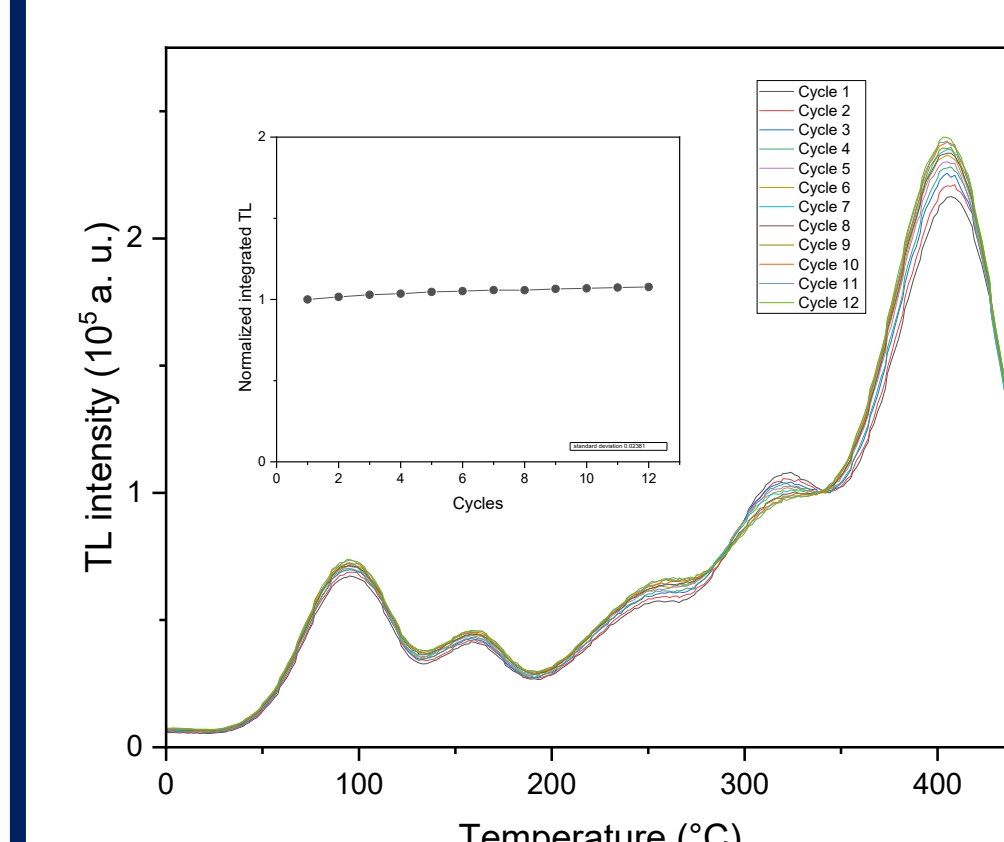


Figure 2. Reproducibility of the glow curves as a function of the number of irradiation-reading cycles of MgO chemically modified with Nd and Li, based on glycine, being exposed to doses of 80 Gy of beta particles prior to each TL reading and the integrated normalized thermoluminescence in function of the irradiation-reading cycles.

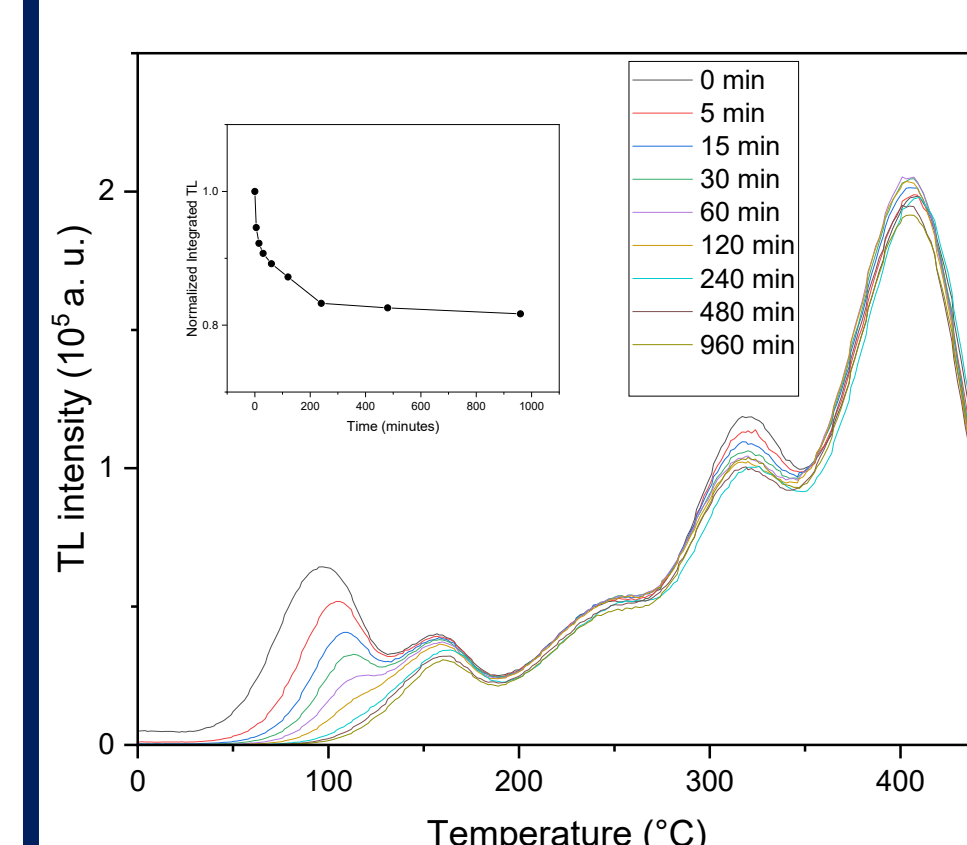


Figure 3. Fading of the thermoluminescent glow curves of MgO powder glycine-based chemically modified with Nd and Li, previously exposed to 80 Gy of beta radiation, and the integrated normalized TL in function of the time.

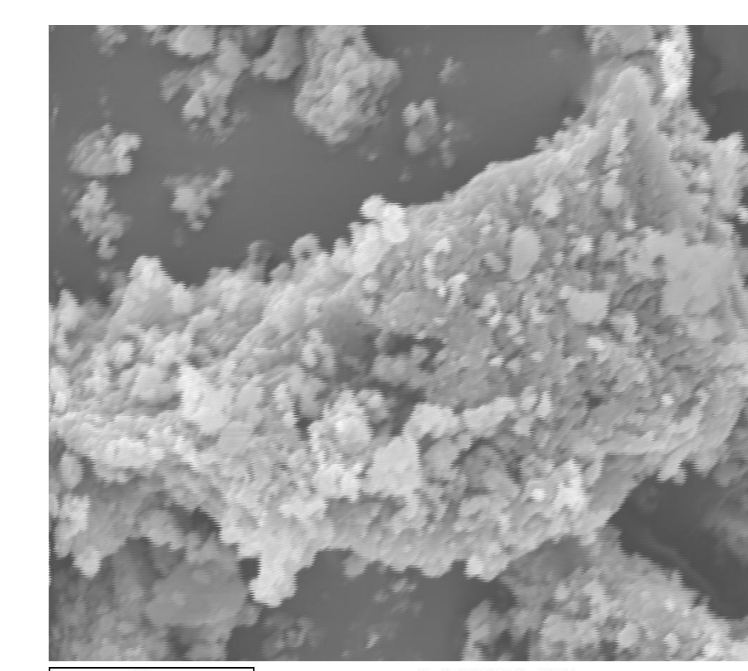


Figure 4. Scanning electron microscopy (SEM) image of a sample of MgO chemically modified with Nd and Li, with magnification of X 3500.

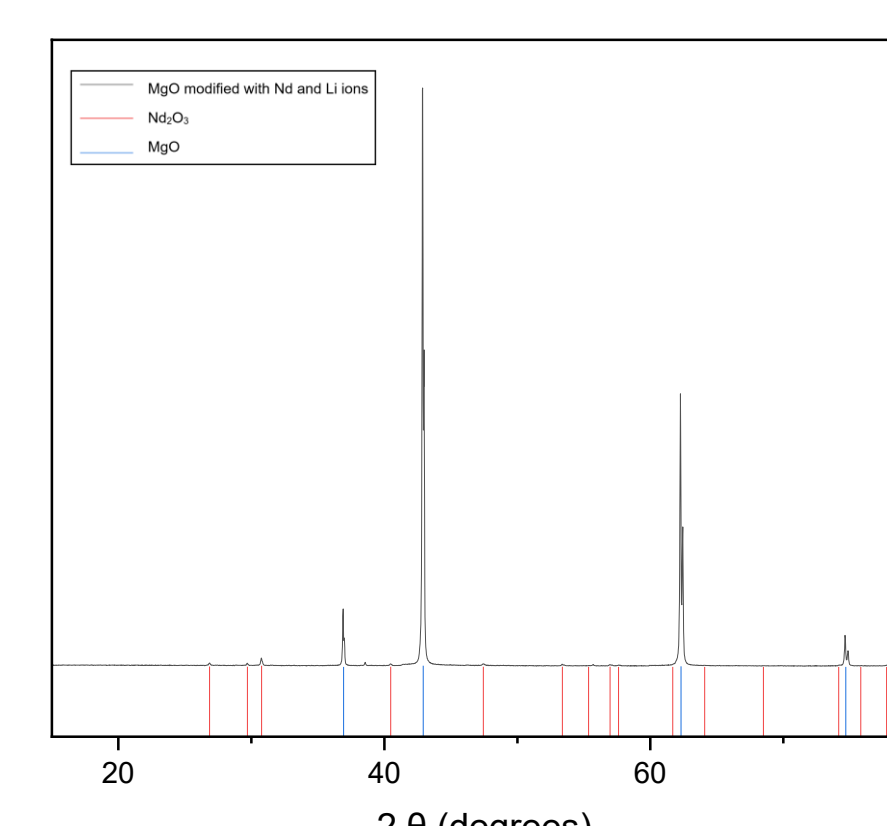


Figure 5. X-ray diffraction of MgO chemically modified with Nd and Li ions (in black), and the file reference 01-077-2179 of *fcc* MgO (in blue), as well as the file reference 01-072-0685 of Nd₂O₃ (in red), from the Inorganic Crystal Structure Database, ICSD.

METHOD	ACTIVATION ENERGY (eV)	FREQUENCY FACTOR (S ⁻¹)
INITIAL RISE	0.31559	2.33E+03
DIFFERENT HEATING RATES	0.506	1.09E+06
LUSHCHIK	0.4173353	7.24E+04
HALPERIN Y BRENER	0.4058379	4.93E+04
Et	0.5027971	1.23E+06
CHEM'S EQUATIONS	0.4073193	5.18E+04
	0.4505023	2.18E+05

Table 1. Different values of the activation energy and the associated frequency factor for each energy, using the highest temperature maximum of TL of MgO chemically modified with Nd and Li.

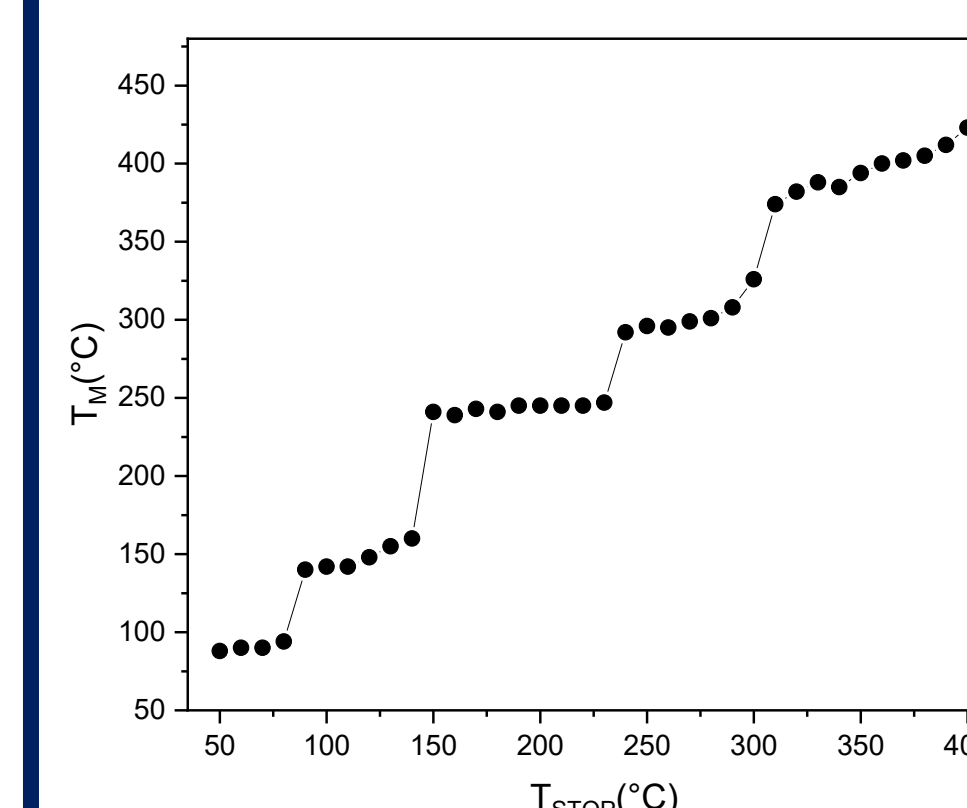


Figure 6. T_M plot as a function of T_{stop} of MgO chemically modified with Nd and Li, obtained from McKeever's gradual heating experiments at a dose of 160 Gy and a heating rate of 2 °C/s.

CONCLUSIONS

It was possible to obtain, through the solution combustion synthesis based on glycine, novel MgO chemically modified with neodymium and lithium phosphors.

The phosphor showed the MgO *fcc* crystalline phase, and also showed the Nd₂O₃ phase, in accordance with the results obtained by X-ray diffraction.

MgO phosphor chemically modified with Nd and Li showed reproducibility and reusability of the thermoluminescent signal. The stability of the aforementioned signal over time was observed.

Different peaks were found using gradual heating experiments, at least ten peaks were found, T_{STOP} region temperature located at 150 °C - 230 °C apparently has at least two peaks with first order kinetics, the other peaks are different of first order kinetics.

The thermoluminescent characteristics obtained in this work for the synthesized materials suggest that they are candidates for TL dosimetry applications of ionizing radiation, for intermediate and high doses.